
Data-Driven Stock Investment Decision Support

A Reinforcement Learning Approach to
Risk-Sensitive Sequential Portfolio Decisions

Master Thesis

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June 1, 2026

Preface

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Abstract

This is temporary abstract text. The purpose of this dummy document is to create enough structure to evaluate the table of contents, chapter ordering, section hierarchy, and later typography choices. The content is placeholder text and should be replaced by the final thesis abstract.

Disclaimer

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Glossary

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Introduction

1.1 Box gallery (visual test)

Definition 1.1.1 (Invariants of unimodular symmetric forms)

Let Z be a finitely-generated free abelian group, and let $q : Z \times Z \rightarrow \mathbb{Z}$ be a unimodular form. To it we assign rank, type (parity), and signature.

1.2 Motivation

Stock investors make sequential decisions under uncertainty. The motivation for this thesis is to explore how reinforcement learning can support such decisions in a transparent and risk-aware way.

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1.3 Problem Formulation

The central problem is how to design a stock investment decision support system that provides meaningful recommendations while avoiding look-ahead bias.

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1.4 Research Questions

The thesis investigates how reinforcement learning, distributional value estimation, and point-in-time data handling can be combined in a decision support system.

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1.5 Scope and Delimitations

The thesis focuses on a proof of concept rather than a production trading platform.

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1.6 Thesis Structure

The thesis is organized into background, related work, methodology, theory, experiments, proof of concept, results, conclusion, and future perspectives.

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Background

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2.1 Financial markets

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2.2 The Investor

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2.3 Portfolio theory

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2.4 Risk, return, and uncertainty

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2.5 Inflation and macroeconomic context

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2.6 Decision support systems in finance

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2.7 Reinforcement learning for investment decisions

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Related Work

3.1 Direct Predecessors

3.2 Closest Application Work

Literature Survey

- 4.1 State of the Art**
- 4.2 Reinforcement Learning for Algorithmic Trading and Portfolio Management**
- 4.3 Distributional Reinforcement Learning**
- 4.4 Uncertainty- and Risk-Sensitive Reinforcement Learning**
- 4.5 Positioning: The Research Gap**
- 4.6 Chosen Solution**
- 4.7 Alternative Solutions**

Theory

- 5.1 Markov Decision Processes
- 5.2 Dynamic Programming
- 5.3 Q-Learning
- 5.4 Deep Q-Networks (DQN)
- 5.5 The Limitation of a Single Scalar Value
- 5.6 Distributional Reinforcement Learning
- 5.7 Quantile Regression and the Quantile Huber Loss
- 5.8 Implicit Quantile Networks (IQN)
- 5.9 Risk-Sensitive Policies via Distortion Measures
- 5.10 Evidential Deep Learning
- 5.11 The Complete Chain

Methodology

This chapter describes how the general theory is shaped into the concrete system and how it is evaluated.

6.1 Research Design

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6.2 Problem Formulation as an MDP

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State space

The state $s_t \in \mathcal{S}$ encodes the information available to the agent at decision time t . It comprises the normalized price window, technical indicators, and the current portfolio allocation, so that $s_t = (\mathbf{p}_{t-w:t}, \mathbf{x}_{t-1})$ where w is the lookback length.

Action space

The action $a_t \in \mathcal{A}$ is the discrete trading decision applied to each asset, drawn from $\{\text{SELL}, \text{HOLD}, \text{BUY}\}$ at fixed position sizes.

Reward design

The reward r_t is the realized log-return of the portfolio over the holding period, optionally adjusted by a risk penalty.

6.3 D-IQN-DSS: The Decision Engine

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6.4 Hierarchical Decision Policy (HDP)

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6.5 Epistemic Uncertainty via EDL

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6.6 The Fusion Rule

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6.7 Risk-Sensitive Decision Scoring

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6.8 Scope and Limitations

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6.9 Experimental Design

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6.10 Evaluation Protocol

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Point-in-time walk-forward backtesting.

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Metrics.

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Acknowledgements

Acknowledgements

First of all, I would like to thank the mathematical community whose documents make LaTeX typography worth studying. This section is included to test the visual spacing of front matter pages.

Additional Material

A.1 Experiment Configuration

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A.2 Additional Tables

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A.3 Box gallery

Definition A.3.1 (Risk-sensitive policy)

A policy π_β is risk-sensitive if it maximizes a distorted expectation $Q_\beta(x, a)$ rather than the plain expectation $\mathbb{E}[Z^\pi(x, a)]$.

Theorem A.3.2 (Bellman optimality)

The optimal action-value function Q^* is the unique fixed point of the Bellman optimality operator $\mathcal{T}$, satisfying $Q^* = \mathcal{T}Q^*$.

Proposition A.3.3 (Monotonicity of the distortion)

If β is a non-decreasing distortion function, then the induced risk measure preserves stochastic dominance between return distributions.

Example A.3.4 (Conditional Value-at-Risk)

Setting $\beta(\tau) = \min(\tau/\eta, 1)$ recovers CVaR_η , which averages the worst η -fraction of outcomes and yields a risk-averse policy.

Remark A.3.5

In practice the quantile samples $\tau \sim U([0, 1])$ are drawn independently for the online and target networks, which decorrelates the temporal-difference error.